

Word problems on volume

Tanks that fill, crates that pack, water that rises, and concrete that has to be paid for.

LESSON 146–149

4 × 40 MIN

MATEMATIKA 6, §31

S7 15 DISTANCE, AREA AND VOLUME

LESSON CLOCK

30'

35'

40'

45'

10

Filling and emptying	30 min
How many fit	35 min
Rising water and displacement	40 min
Volume, mass and cost	45 min
Homework	10 min

LEARNING OBJECTIVES

- Find a filling or emptying time from a capacity and a rate.
- Decide how many small solids pack into a larger one.
- Use the rise in a water level to find a volume, and the reverse.
- Turn a volume into a mass or a cost with a rate per unit volume.

TEXTBOOK

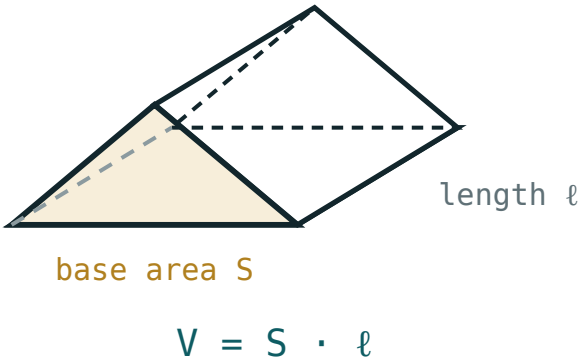
National	pp. 424–435
Cambridge	Stage 7, pp. 162–170
Workbook	Exercise 15.4–15.5

01

Explanation

Filling and emptying

Every one of these problems has the same two steps: find the volume in litres, then divide by the rate. The work is in the units, not the arithmetic.



Base area times height — the volume every tank problem starts from

TANK	CAPACITY	RATE	TIME
$60 \times 40 \times 50\text{ cm}$	120 litres	8 l/min	15 min
$1\text{ m} \times 80 \times 50\text{ cm}$	400 litres	10 l/min	40 min
a 200-litre drum	200 litres	5 l/min	40 min
a 90-litre tank draining	90 litres	3 l/min	30 min

THE METRE IN THE SECOND ROW IS THE TRAP

1 m is 100 cm, so the volume is $100 \cdot 80 \cdot 50 = 400\,000\text{ cm}^3$. Leaving the 1 in place gives 4000 cm^3 — four litres, for a tank a person could sit in.

How many fit

When the small solid divides the large one exactly along each edge, count along each edge and multiply the three counts.

CONTAINER	ITEM	ALONG EACH EDGE	HOW MANY
$40 \times 32 \times 30\text{ cm}$	$10 \times 8 \times 5\text{ cm}$	4, 4, 6	96
cube of edge 12 cm	cube of edge 3 cm	4, 4, 4	64
$10 \times 8 \times 6\text{ cm}$	cube of edge 2 cm	5, 4, 3	60
$1\text{ m} \times 40 \times 24\text{ cm}$	$20 \times 10 \times 8\text{ cm}$	5, 4, 3	60

DIVIDING THE VOLUMES IS ONLY AN UPPER LIMIT

$38\,400 \div 400 = 96$ happens to be right in the first row because the boxes fit along every edge. If they did not, the volume division would still give 96 and the true answer would be smaller — so count along the edges, and use the division only as a check.

Rising water and displacement

Water in a tank is a cuboid whose base is the tank's base and whose height is the depth. Two questions follow at once.

$$\text{volume of water} = \text{base area} \times \text{depth so depth} = \frac{\text{volume}}{\text{base area}}$$

QUESTION	WORKING	ANSWER
water 6 cm deep on a 40×25 cm base	$1000 \cdot 6$	6 litres
3 litres poured onto a 30×20 cm base	$3000 \div 600$	5 cm deep
a stone raises 6 cm to 7.5 cm on a 40×25 base	$1000 \cdot 1.5$	1500 cm^3 of stone
a cube of edge 5 cm sunk in a 25×20 tank	$125 \div 500$	a rise of 0.25 cm

A SUNKEN OBJECT TAKES EXACTLY ITS OWN VOLUME OF ROOM

That is why the rise in level, multiplied by the base area, gives the volume of an object of any shape at all — the oldest measuring trick in mathematics.

Volume, mass and cost

A rate per unit volume turns the volume into whatever the question is really about.

OBJECT	VOLUME	RATE	ANSWER
an iron block $10 \times 5 \times 2$ cm	100 cm^3	7.8 g per cm^3	780 g
a full 120-litre tank of water	120 litres	1 kg per litre	120 kg
a concrete slab $4 \times 3 \times 0.1$ m	1.2 m^3	2.4 t per m^3	2.88 t
sand for a pit $2 \times 1.5 \times 0.4$ m	1.2 m^3	90 000 so‘m per m^3	108 000 so‘m

MATCH THE UNIT OF THE RATE TO THE UNIT OF THE VOLUME

A rate in g per cm^3 needs the volume in cm^3 ; a rate per m^3 needs m^3 . Converting the volume first is always safer than trying to convert the rate.

Mixed problems

Longer questions simply chain these steps together.

PROBLEM	STEPS	ANSWER
a $50 \times 40 \times 30$ cm tank filled to two-thirds	$60 \cdot \frac{2}{3}$	40 litres
two taps, 6 and 4 l/min, on a 120-litre tank	$120 \div 10$	12 min
a 300-litre tank filling at 12 and leaking at 2	$300 \div 10$	30 min
a 120-litre tank half full, topped up at 4 l/min	$60 \div 4$	15 min

TWO TAPS TOGETHER ADD THEIR RATES

Exactly as two people working together add their rates of work. A leak subtracts, which is the same idea with a sign.

02

Worked examples

EXAMPLE 1 A tank measures 60 cm by 40 cm by 50 cm. A tap delivers 8 litres a minute. How long does it take to fill?

1

$60 \cdot 40 \cdot 50 = 120\,000\text{ cm}^3.$

2

That is 120 litres.

3

$120 \div 8 = 15\text{ minutes.}$
Check: $8 \cdot 15 = 120\checkmark$

ANSWER

15 minutes

EXAMPLE 2 How many boxes 10 cm by 8 cm by 5 cm fit into a crate 40 cm by 32 cm by 30 cm?

① Along the edges: $40 \div 10 = 4$, $32 \div 8 = 4$, $30 \div 5 = 6$.

② $4 \cdot 4 \cdot 6$.

③ = 96 boxes.

Volume check: $38\,400 \div 400 = 96 \checkmark$

ANSWER

96 boxes

EXAMPLE 3 A tank with base 40 cm by 25 cm holds water 6 cm deep. A stone is lowered in and the level rises to 7.5 cm. Find the volume of the stone.

① Base area $40 \cdot 25 = 1000 \text{ cm}^2$.

② The rise is 1.5 cm.

③ $1000 \cdot 1.5 = 1500 \text{ cm}^3$.

The stone takes exactly its own volume of room.

ANSWER

1500 cm³

03 Interactive model

Do the displacement question with a real jar and a stone; the number on the board becomes a measurement the class has just watched being made.

Tanks, crates and stones

A $60 \times 40 \times 50$ cm tank holds:

12 litres

120 litres

1200 litres

120 000 litres

Filling it at 8 l/min takes:

12

15

20

960

A 1 m by 80 cm by 50 cm tank holds:

4 litres

40 litres

400 litres

4000 litres

Cubes of edge 3 cm in a cube of edge 12 cm:

4

16

64

48

3 litres on a 30×20 cm base is deep:

5 cm

50 cm

0.5 cm

6 cm

A rise of 1.5 cm on a 1000 cm^2 base means an object of:

150 cm^3

1500 cm^3

15 cm^3

666 cm^3

Concrete at 2.4 t per m^3 in a 1.2 m^3 slab weighs:

2 t

2.88 t

3.6 t

0.5 t

Taps of 6 and 4 l/min fill 120 litres in:

10

12

20

24

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Capacity	Sig'im	Вместимость
To fill	To'ldirmoq	Наполнять
To empty, to drain	Bo'shatmoq	Опорожнять
Rate of flow	Oqim tezligi	Скорость потока
Water level	Suv sathi	Уровень воды
Displacement	Siqib chiqarish	Вытеснение
Mass	Massa	Масса
Per cubic metre	Bir kub metr uchun	За кубический метр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A capacity in litres comes from a volume in:

cm², divided by 1000

cm³, divided by 1000

cm, times 1000

m³, times 100

To find a filling time you:

multiply capacity by rate

divide capacity by rate

add them

divide rate by capacity

The number of small boxes that fit is found by:

dividing the volumes only

counting along each edge

adding the edges

guessing

A sunken object raises the water by:

its surface area

its own volume ÷ the base area

its mass

its height

A rate of 7.8 g per cm^3 needs the volume in:

m^3

litres

cm^3

cm^2

A tank filling at 12 and leaking at 2 l/min gains:

14

10

6

24

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The capacity of a $60 \times 40 \times 50$ cm tank

ANSWER 120 litres

2. A 120-litre tank filled at 8 l/min

ANSWER 15 min

3. A 200-litre drum filled at 5 l/min

ANSWER 40 min

4. A 90-litre tank draining at 3 l/min

ANSWER 30 min

5. Cubes of edge 3 cm in a cube of edge 12 cm

ANSWER 64

6. Water 6 cm deep on a 40×25 cm base

ANSWER 6 litres

7. The mass of 1 litre of water

ANSWER 1 kg

Medium · the standard question

7 PROBLEMS

1. The capacity of a 1 m by 80 cm by 50 cm tank

ANSWER 400 litres

2. ...filled at 10 l/min

ANSWER 40 min

3. Boxes $10 \times 8 \times 5$ cm in a crate $40 \times 32 \times 30$ cm

ANSWER 96

4. Cubes of edge 2 cm in a $10 \times 8 \times 6$ cm box

ANSWER 60

5. 3 litres poured onto a 30×20 cm base: the depth

ANSWER 5 cm

6. An iron block $10 \times 5 \times 2$ cm at 7.8 g per cm^3

ANSWER 780 g

7. The water in a full 120-litre tank

ANSWER 120 kg

Hard · stretch and reason

7 PROBLEMS

1. A stone raises 6 cm to 7.5 cm on a 40×25 base: its volume

ANSWER 1500 cm³

2. A cube of edge 5 cm sunk in a 25×20 cm tank: the rise

ANSWER 0.25 cm

3. A concrete slab 4 m by 3 m by 0.1 m at 2.4 t per m³

ANSWER 2.88 tonnes

4. Sand at 90 000 so'm per m³ for a pit $2 \times 1.5 \times 0.4$ m

ANSWER 108 000 so'm

5. A $50 \times 40 \times 30$ cm tank filled to two-thirds

ANSWER 40 litres

6. Taps of 6 and 4 l/min on a 120-litre tank

ANSWER 12 min

7. A 300-litre tank filling at 12 and leaking at 2 l/min

ANSWER 30 min

07 Homework

Homework — 5 tasks

Convert every length to one unit before the first multiplication, and write the unit on every line.

1. A tank is 70 cm by 50 cm by 40 cm. Find its capacity in litres.
2. That tank is filled by a tap delivering 7 litres a minute. How long does it take?
3. How many cubes of edge 4 cm fit into a box 20 cm by 16 cm by 12 cm?
4. 4.5 litres of water is poured into a tank with base 45 cm by 20 cm. How deep is the water?
5. Concrete weighs 2.4 tonnes per m³. Find the mass of a slab 5 m by 2 m by 0.2 m.